Krithika Anand

M.Sc. Chemical Biotechnology

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LinkedIn



PROFILE

M.Sc. Chemical Biotechnology student at TUM with a strong hands-on experience in molecular biology and protein engineering. I excel at designing and carrying out experiments, troubleshooting challenges, and delivering reliable results. I am now looking for a challenging role that integrates my scientific expertise to deliver impactful projects.

EDUCATION

04/2024 – Present
Straubing, Germany

M.Sc. Chemical Biotechnology
Technical University of Munich, Germany

08/2019 – 04/2023
Chennai, India

B. Tech. Biotechnology
Rajalakshmi Engineering College

EXPERIENCE

11/2025 – Present
Munich, Germany

Research Assistant in (WSSB)
Technical University of Munich (TUM), Campus Garching

- Performed routine laboratory tasks including sample preparation, media preparation, and maintaining sterile working conditions.
- Engineered a terpene synthase enzyme to improve thermostability through experimental design and testing.
- Designed and introduced targeted mutations using PCR-based site-directed mutagenesis.
- Performed cloning, plasmid amplification, and sequence verification in *E. coli*.
- Conducted functional expression and screening in *E. coli*.
- Analyzed terpene production using gas chromatography–mass spectrometry (GC–MS), identifying improved variants.
- Performed troubleshooting to resolve experimental issues and improve reproducibility.
- Established expression systems using pACYC and pET-SUMO vectors for functional assays and protein studies.
- Performed protein overexpression (IPTG induction) and purification using FPLC.
- Evaluated protein stability using differential scanning calorimetry (DSC).

03/2025 – 06/2025
Straubing, Germany

Research Assistant in (BFM)
Technical University of Munich, Campus Straubing

- Performed thermostability analysis of fluorescent proteins (DsRed, EosFP, mGL, mYFP, mKO).
- Designed and executed experiments using additives (glucose, sorbitol, trehalose, imidazole) and polymer matrices (maltodextrin, hydroxypropyl cellulose).
- Prepared protein samples and quantified concentrations via spectrophotometry.
- Conducted thermal stability assays using qPCR-based fluorescence measurements.

- Maintained detailed and structured documentation of experimental procedures and results.
- Analyzed data to identify optimal conditions for enhanced protein stability.

01/2023 – 03/2023
Chennai, India

Research Project Assistant
University of Madras, Campus Guindy
- Isolated PBMCs and performed cell-based assays to evaluate vaccine-induced immune responses.
- Conducted peptide stimulation experiments to assess T-cell activation and specificity.
- Performed ELISA to quantify pro-inflammatory cytokines in stimulated samples.
- Carried out peptide titration to optimize experimental conditions for immune response assays.
- Measured oxidative stress and protein damage markers using biochemical assays.
- Analyzed data using GraphPad Prism to evaluate immune and inflammatory responses.

01/2021 – 02/2021
Chennai, India

Student Intern
MGM Healthcare
- Performed basic microbiology techniques including microbial isolation, culturing, and staining.
- Processed and tested patient samples to isolate and identify bacterial pathogens.
- Assisted in routine reporting of test results for clinical diagnosis.
- Assisted in laboratory setup, sterilization, media preparation and aseptic handling procedures.

SKILLS

- **Sample & media preparation**
- **Microbial culture handling & aseptic techniques**
- **Molecular cloning & site-directed mutagenesis**
- **Analytical techniques (GC–MS, ELISA, western blot, spectrophotometry)**
- **Cell-based assays & microbiological techniques**
- **Recombinant protein expression & purification (FPLC)**
- **Experimental workflow execution**
- **Documentation & data recording**
- **Data analysis & visualization (GraphPad Prism, Google Colab, MS Excel)**

PUBLICATIONS

01/2023

Strategies for enhancing the efficacy of anaerobic digestion of food industry wastewater: An insight into bioreactor types, challenges, and future scope.
Chemosphere

LANGUAGES

German	English	Tamil	Telugu
A1	C2	C2	(native)